

MODULAR FORMS MMATH II YEAR

26 NOVEMBER 2025
DURATION 3 HOURS
FINAL EXAM 2025
MAX MARKS: 90
ANSWER ALL QUESTIONS

You are allowed one page (both sides) cheat sheet

1. Let $\eta(z)$ denote the Dedekind eta function. For $q = 2, 3, 5, 7, 13$ let $r = 24/(q - 1)$ and define

$$\Phi(z) = \left(\frac{\eta(qz)}{\eta(z)} \right)^r.$$

(a) State Dedekind's functional equation.

(b) Assuming Dedekind's functional equation and required properties of the Dedekind sum, show that $\Phi(z)$ is a modular function for $\Gamma_0(q)$.

(c) Show that Φ has a simple zero at infinity.

(5 + 18 + 7 = 30 marks)

2. Let $\Gamma = SL_2(\mathbb{Z})$. Let $f \in M_{2k}(\Gamma)$ and $T(m)$ be the m -th Hecke operator. Suppose the Fourier expansions are given by

$$f(z) = \sum_{n=0}^{\infty} a(n) e^{2\pi i n z}, \quad \text{and} \quad (T(m)f)(z) = \sum_{n=0}^{\infty} A(n) e^{2\pi i n z}$$

(a) Show that

$$A(n) = \sum_{\substack{d|(m,n) \\ d \geq 1}} d^{2k-1} a\left(\frac{mn}{d^2}\right).$$

(b) Let $\tau(n)$ denote the Ramanujan tau function. Show that for any prime p , either $\tau(p) \neq 0$ or $\tau(p^2) \neq 0$.

(10 + 10 = 20 marks)

3. Let $T_k(m)$ be the m -th Hecke operator and for any prime $q|N$ with $Q = q^\ell || N$, let w_Q be the Atkin-Lehner operator on $S_k(\Gamma_0(N))$.

(a) For $(m, N) = 1$, show that

$$T_k(m)w_Q = w_Q T_k(m).$$

(b) For $(m, N) \neq 1$, show that $T_k(m)$ may not be self adjoint.

(c) Suppose $q^2|N$, then show that $T_k(q)f \in S_k^{\text{old}}(\Gamma_0(N))$ for any $f \in S_k(\Gamma_0(N))$.

(10+10+10=30 marks)

4. Let

$$f(z) = \sum_{n=1}^{\infty} a_f(n) e(nz) \in S_k(\Gamma)$$

be a Hecke form for the full modular group.

(a) Show that

$$g_1(z) := \sum_{n=1}^{\infty} a_f(6n) e(nz)$$

is a modular form of some level.

(b) Show that

$$g_2(z) := \sum_{\substack{n=1 \\ (n,6)=1}}^{\infty} a_f(n) e(nz)$$

is a modular form of level 36.

(10+10=20 marks)

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$$

$$\mathbb{Q}, \mathbb{R}, \mathbb{C}$$

$$\begin{matrix} 5 \\ 2 \end{matrix}$$

$$\begin{pmatrix} p & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} a & a & 0 & 1 \\ , & & & \end{pmatrix}$$

$$= \begin{pmatrix}$$